

WildfireIQ Kamloops

An AI-Powered Wildfire Risk, Air Quality and Community Preparedness Platform for the Thompson-Okanagan Region

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Research Project: Goals and Objectives

Overview. Wildfires in British Columbia are growing in frequency and scale, and the Thompson-Okanagan region around Kamloops has been among the most severely affected parts of Canada over the past decade. Large amounts of publicly available information exist on wildfire activity, local weather, and air quality, yet no single tool brings this information together in a way that is useful to the people most affected. WildfireIQ Kamloops is designed to fill that gap.

The project will produce a publicly accessible web platform with four connected features: a map showing predicted daily wildfire risk levels across the region; a monitor tracking current air quality and forecasting smoke conditions 48 hours ahead; a personalized preparedness guide helping individuals take concrete steps to protect themselves before and during fire season; and a trend analysis tool showing how fire seasons and regional climate have changed over 30 years. All data used is freely and publicly available. The project involves no research on human subjects.

Short-Term Goals (Months 1–7).

- Gather and organize all required public datasets: BC Wildfire Service fire records back to 1999, daily Kamloops weather data, Canada-wide fire weather index values, NASA satellite fire detection feeds, air quality readings from the BC Air Data Archive, wildfire smoke forecast data, BC Emergency Management evacuation zone maps, and 30-year regional climate projections from ClimateData.ca.
- Develop and test two predictive models: one that estimates the daily likelihood of wildfire activity by geographic area, and one that forecasts air quality in Kamloops up to 48 hours ahead based on nearby fire activity and wind patterns.
- Build and launch the first two platform features, the Wildfire Risk Map and the Air Quality Monitor, connected to live data.

Long-Term Goals (Months 8–12).

- Complete the Community Preparedness Hub with personalized FireSmart checklists, live evacuation zone status by neighbourhood, and a points and achievement system to encourage follow-through.
- Deliver the Climate Trend module showing how Thompson-Okanagan fire seasons have changed over 30 years alongside projected temperature and precipitation shifts. An optional TRU campus carbon progress layer will be added if institutional data becomes available.

- Test both predictive models against held-out data from recent fire seasons and document accuracy results.
- Submit the final research report to TRUSpace, present at a TRU research event, and release all code publicly for others to build on.

Project Description

Justification

BC's 2023 wildfire season was the largest on record, burning over 2.8 million hectares. The Thompson-Okanagan corridor around Kamloops has seen repeated large-scale evacuations in recent years, with serious consequences for residents, local businesses, and public health. Wildfire smoke has been linked to increased emergency room visits, school and workplace closures, and lasting respiratory harm in vulnerable populations including children, the elderly, and outdoor workers.

Despite the severity and regularity of these events, the tools available to ordinary people remain scattered and incomplete. Air quality apps do not connect to fire risk data, evacuation resources are static, and no platform synthesizes the available information into clear, actionable guidance. This project directly addresses that gap.

The research addresses all four of TRU's sustainability pillars. Environmentally, it produces original research linking historical fire patterns, daily weather, and satellite data to forward-looking risk assessments for this specific region. Socially, it delivers a practical tool for TRU students and Kamloops residents who face real risk each summer. Economically, it models how smoke exposure drives healthcare demand, which is relevant to regional planning decisions. Culturally, the platform is built to incorporate Traditional Ecological Knowledge on fire management as those datasets become available through the Wildfire Resilience Consortium of Canada, of which TRU is a foundational partner.

Methodology

Phase 1: Data Collection and Predictive Modelling. All data sources are freely available from government and research agencies. The wildfire risk model will learn from 25 years of historical fire records combined with corresponding weather conditions, fire weather index values, and satellite-measured vegetation health, producing a daily risk rating for geographic zones across the Thompson-Okanagan expressed as Low, Moderate, High, or Extreme. The air quality forecasting model will learn the relationship between nearby fire activity, wind conditions, and observed air quality readings in Kamloops, then apply that relationship to produce 48-hour forecasts. Both models will be evaluated against data from the 2022 and 2023 fire seasons held back from training, so results reflect genuine predictive ability.

Phase 2: Platform Development. The four features will be built and connected to live data and model outputs. The Wildfire Risk Map will show an interactive regional map with colour-coded daily risk zones overlaid with current and historical fire locations. The Air Quality Monitor will display the current air quality index for Kamloops, the 48-hour forecast, and a calendar of past smoke events linked to nearby fires. The Community Preparedness Hub will take a user's neighbourhood as input, generate a tailored FireSmart action checklist, display active evacuation orders for their area, and track completed actions through a points system. The Climate Trend module will present a 30-year chart of fire season

severity for the Thompson-Okanagan alongside projected regional temperature and precipitation changes.

Anticipated Outcomes

- A fully operational, publicly accessible platform with all four features live, documented, and available to the Kamloops community and beyond.
- Two trained predictive models with complete methodology descriptions and accuracy results reported transparently and reproducibly.
- A final research report published to TRUSpace and a fully open codebase available to other researchers, students, and community organizations who wish to extend or adapt the work.

Knowledge Dissemination

Research findings will be shared through three channels. The final report will be submitted to TRUSpace as required. The platform will remain publicly accessible throughout and beyond the grant period as a live research output that continues to serve the community. A presentation and poster will be delivered at a TRU student research event or sustainability symposium. Given the direct alignment between this project and TRU’s active role in the Wildfire Resilience Consortium of Canada, findings will also be shared proactively with the TRU Sustainability Office for potential connection with existing institutional initiatives.

Proposed Budget, Justification, and Timeline

Budget Summary

Category	Amount	Justification
Personal Stipend	\$400	Researcher time for data acquisition, model development, platform construction, testing, and documentation over the 12-month project period, allocated at the maximum permitted rate of 16% of the total award.
Cloud Hosting and Computing	\$200	Server hosting for the live platform and predictive models, covering 12 months of backend and frontend hosting and compute time for model training.
Mapping and AI Services	\$300	Third-party API and service costs for interactive geospatial rendering and AI-assisted content generation within the platform, including mapping services for the Wildfire Risk Map and language model API access for generating dynamic, personalized outputs in the Community Preparedness Hub.
Conference and Dissemination	\$1,000	Costs associated with presenting and disseminating research findings, including registration fees and related travel expenses for attendance at one or more regional or national conferences in sustainability, environmental informatics, or computing science, as well as any associated materials costs for presenting at TRU research events.

Category	Amount	Justification
Software and Storage	\$350	External hard drive for secure dataset backup (\$120); research reference materials (\$130); code repository hosting during active development (\$100).
Contingency	\$250	Reserve for unanticipated project costs, including hosting or infrastructure overruns, unforeseen software or service requirements, or additional dissemination expenses. Any unused funds will be returned.
Total	\$2,500	

All expenditures beyond the personal stipend will be reconciled against itemized receipts and submitted to the faculty supervisor. Any budget category change exceeding 20% of the stated amount will be approved in writing by the TRU Sustainability Office prior to the expenditure.

Project Timeline

Month(s)	Milestones
1 – 2	All datasets acquired, cleaned, and organized. Development environment established and project repository initialized.
3 – 4	Wildfire risk prediction model trained, tested, and documented. Air quality forecasting model trained, tested, and documented.
5 – 6	Wildfire Risk Map and Air Quality Monitor built and connected to live data and model outputs. Platform deployed and publicly accessible online.
7 – 8	Community Preparedness Hub completed: personalized checklists, evacuation zone integration, and progress tracking system fully operational.
9 – 10	Climate Trend module completed. All four features integrated, tested together, and refined based on review.
11	Findings presented at conference or TRU research event. Platform publicly launched. Full codebase released as open source.
12	Final research report submitted to TRUSpace. All receipts reconciled and submitted to faculty supervisor. Project formally closed.

Ethics and Data Statement

This project draws exclusively on publicly available datasets published by government agencies and research institutions, including BC Wildfire Service, Environment Canada, NASA, BC Emergency Management, Interior Health, and ClimateData.ca. No personal data is collected at any stage. No interviews, surveys, or observations involving human participants are conducted. Ethics approval is therefore not required. The platform itself will not store any personally identifiable information about users.